

Ecolight Maestro

User Manual

1. Introduction

This manual explains how to operate Ecolight Maestro, an automatic street light controller built on the LPC2129 ARM7 microcontroller. It covers system startup, normal automatic operation, and how to manually update the Real-Time Clock (RTC) using the keypad menu.

2. System Overview

On power-up, the system initializes the LCD, RTC, ADC, and GPIO peripherals, then enters continuous operation, reading the time and light level (via the LDR/ADC) and driving the street light LEDs automatically.

- LCD 16x2: shows current time, date, day, ADC value, and system status
- 4x4 Keypad: used to enter and navigate the RTC configuration menu
- EINT0 push button (P0.16): opens the RTC configuration menu on demand
- LEDs: simulate the street lights

3. Normal (Automatic) Operation

No user action is required for day-to-day use. The controller checks two conditions continuously:

- LED OFF: Time is before 18:00, OR the ADC (light) reading is above 300 (bright conditions)
- LED ON: Time is 18:00 or later, AND the ADC reading is 300 or below (dark conditions)

The LCD updates automatically to reflect the current time, date, and light status.

4. Updating the Time and Date (RTC Menu)

Follow these steps to change the hour, minute, second, day, date, month, or year:

1. Press the EINT0 push button (P0.16). This interrupts normal operation and opens the RTC configuration menu on the LCD.
2. The LCD displays a numbered list of menu options (see table below).
3. On the keypad, press the number key corresponding to the parameter you want to change.
4. Enter the new value using the keypad when prompted.
5. The system saves the updated value to the RTC and returns to the menu.
6. Press 8 to exit the menu and resume normal automatic operation.

Menu Reference

Key	Action
1	Update Hour

2	Update Minute
3	Update Second
4	Update Day
5	Update Date
6	Update Month
7	Update Year
8	Exit Menu

5. Troubleshooting

LCD shows nothing / blank

- Check power supply and LCD contrast potentiometer
- Verify LCD data/control lines match the pin configuration used in the firmware

LEDs never turn ON at night

- Confirm the RTC time is set correctly using the menu (Section 4)
- Check the LDR wiring and voltage divider — the ADC reading should drop in low light
- Verify the ADC threshold (300) still matches your hardware's light levels; recalibrate if needed

Keypad / menu not responding

- Confirm the EINT0 push button is wired to P0.16 and is triggering the interrupt
- Check keypad row/column wiring against the KPM pin configuration

Time resets after power loss

- This is expected if the RTC has no backup supply; re-enter the time via the menu after power-up

6. Safety Notes

- Disconnect power before changing any wiring
- Use current-limiting resistors on all LEDs to prevent damage to GPIO pins
- Double-check voltage levels before connecting to any actual (non-simulated) street light driver circuit